

Reminder.

I ideal $\Leftrightarrow 0 \in I, I+I \subseteq I, R \cdot I \subseteq I$.

$\langle f_1, \dots, f_m \rangle$ ideal generated by $f_1, \dots, f_m \in R$, i.e. $f_1 R + \dots + f_m R$

$I(X) := \{ f \in k[x_1, \dots, x_n] : f = 0 \text{ on } X \}$.

1.5 Ideals in the ring of polynomials in one variable.

1.5.1. Def. For a polynomial $f = a_0 x^m + a_1 x^{m-1} + \dots + a_{m-1} \in k[x] \setminus \{0\}$

with $m \in \mathbb{N}_0$ and $a_0 \neq 0$, we call $LT(f) = a_0 x^m$ the leading term of f .

1.5.2. Prop. For every $f \in k[x]$ and every $g \in k[x] \setminus \{0\}$, there exists a unique representation $f = q \cdot g + r$ with $q, r \in k[x]$ and $\deg r < \deg g$.

Moreover, there is an algorithm that calculates q and r from f and g .

Proof: We show the existence by providing an algorithm that works. The algorithm that we devise starts with $q=0, r=f$, which satisfy $f = q \cdot g + r$, and iteratively works towards fulfilling the condition $\deg r < \deg g$, while maintaining this equality.

We can take r and replace it with $r - \frac{LT(r)}{LT(g)}g$ (2.)
whenever $\deg r \geq \deg g$. But to maintain the equality,
we then need to change g to $g + \frac{LT(r)}{LT(g)}$. Let's check.

$$f = (q+T) \cdot g + (r - T \cdot g)$$

holds for $T = \frac{LT(r)}{LT(g)}$. So, $q+T$ is a new q and
 $r - T \cdot g$ is a new r . This update keeps the equality
 $f = q \cdot g + r$ satisfied. With each such update, $\deg r$
gets smaller. So, eventually we arrive at $\deg r < \deg g$.

As a pseudocode:

Univariate-Division(f, g):

$r := f$ $\triangleright$ remainder in progress

$q := 0$ $\triangleright$ quotient in progress

while $\deg r \geq \deg g$:

$\triangleright$ Invariant: $f = q \cdot g + r$ $\leftarrow$ this will stay true in each iteration, by design.

$T := \frac{LT(r)}{LT(g)}$ $\triangleright$ we in fact get a term because $\deg r \geq \deg g$

$r := r - T \cdot g$ $\triangleright$ the degree drops, because the leading terms cancel out

$q := q + T$ $\triangleright$ new term added to the quotient.

return q, r

This is the standard long division algorithm for univariate polynomials, whose correctness has been discussed above. It remains to show that the choice of q and r is unique.

(3)

Assume, we have $f = q \cdot g + r = Q \cdot g + R$ with (4)
 $q, Q, r, R \in k[x]$ and $\deg r < \deg g, \deg R < \deg g$.

If $q = Q$, then $r = R$, because $r = f - q \cdot g = f - Q \cdot g = R$.
So, assume $q \neq Q$. Then $\underbrace{(q - Q)}_{\neq 0} \cdot g = \underbrace{R - r}_{\deg < \deg g}$ gives

a contradiction, because the l.h.s has $\deg \geq \deg g$ and the
r.h.s has degree $< \deg g$. This gives uniqueness of q, r . $\square$

Corollary 15.3. Every polynomial $f \in k[x] \setminus \{0\}$ has at most
 $\deg f$ zeros in k .

Proof. Exercise. If $r \in k$ and $f(r) = 0$, division of f by $x - r$ gives
the remainder $f(r) = 0$. $\square$

Corollary 1.5.4 Let I be an ideal in $k[x]$. Then I can be generated by just one polynomial, i.e. $I = \langle g \rangle$ for some $g \in k[x]$. (5)

Proof: If $I = \{0\}$, we just take $g = 0$. If $I \neq \{0\}$, Pick $g \in I \setminus \{0\}$ of smallest degree. Let's check $I = \langle g \rangle$. Since $g \in I$, we have $\langle g \rangle \subseteq I$. Let's check $I \subseteq \langle g \rangle$. Take any $f \in I$ and divide it by g via the long division. One obtains $f = q \cdot g + r$ with $q, r \in k[x]$ and $\deg r < \deg g$. But then $r = \underbrace{f}_{\in I} - q \cdot \underbrace{g}_{\in I} \in I$. So r is in the ideal and $\deg r < \deg g$. But g was chosen to be a non-zero polynomial in I of smallest degree. Hence, r has to be 0. $\Rightarrow f = q \cdot g \in \langle g \rangle$. $\square$

Def 1.5.5. Ideals generated by just one element are called principal. Rings with all ideals being principal are called principal ideal domains (PID).

$k[x]$ is an example of PID, $\mathbb{Z}$ is another PID. Both interesting for CS (coding theory, cryptography!)

Question: $f_1, \dots, f_m \in k[x] \Rightarrow I = \langle f_1, \dots, f_m \rangle$

How do we actually find $g \in k[x]$ with $I = \langle g \rangle$.

Def 15.6. A greatest common divisor of f and g in $k[x]$ is a polynomial $h \in k[x]$ with the following properties:

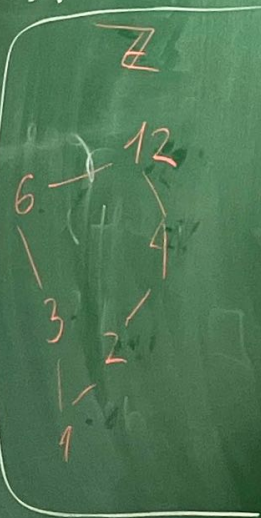
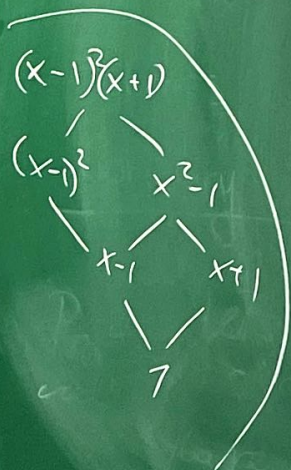
(i) h divides both f and g

(ii) if $p \in k[x]$ divides f and g , then p divides h .

The notation for this h is $\gcd(f, g)$,

which is somewhat ambiguous, because

if h is a gcd, then $c \cdot h$ with $c \in k \setminus \{0\}$ is also a gcd.

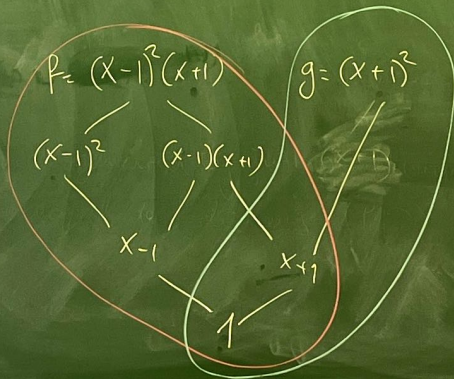
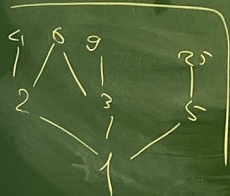


15.7 Prop. Let $f, g \in k[x]$. Then the following are true:

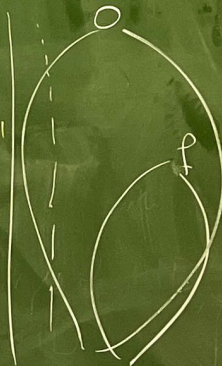
$$\langle f, g \rangle = \langle \gcd(f, g) \rangle \quad (7)$$

- (i) $\gcd(f, g)$ exists and is unique up to a multiplication with an element of $k \setminus \{0\}$.
- (ii) $\langle f, g \rangle = \langle \gcd(f, g) \rangle$
- (iii) There exists an algorithm that computes $\gcd(f, g)$ for given f and g .

Proof: We first address a degenerate case $\gcd(f, g) = 0$. Clearly, $\gcd(f, g) = 0 \Leftrightarrow f = g = 0$.



$$\gcd(f, g) = x+1$$



h divisor of f
 $\Leftrightarrow hq = f$ for some $q \in k[x]$.

h divisor of 0
 $\Leftrightarrow h \cdot q = 0$ for some q .

Divisors of 0 — the whole $k[x]$.

0 divisor of $f \Leftrightarrow 0 \cdot q = f$ for some q .

Only $f=0$ has 0 as a divisor.